

# Real products. Real budgets. A room that actually fits.

## CORE APPROACH

An AI-assisted bathroom designer that turns a plain-language vision, a budget, and a room size into a coordinated bundle of real KOHLER products -- laid out to true scale in the actual room, with every spatial trade-off reported honestly instead of hidden.

- **Deterministic, not a black box.**  
Theme matching and product scoring are explainable weighted functions over the real catalog -- reproducible, auditable, no external LLM call in the recommendation path.
- **Physically honest layout.**  
A functional-station placement engine runs the fixture assembly along the room's actual longer wall, respects door-swing clearance, and reports "room is tight" instead of silently overlapping fixtures.
- **Built on the real catalog.**  
~1,500 real KOHLER SKUs with real prices, categories, and product pages -- not placeholder data.

**~1,500**

real KOHLER SKUs in the catalog

**12**

product categories supported

**25**

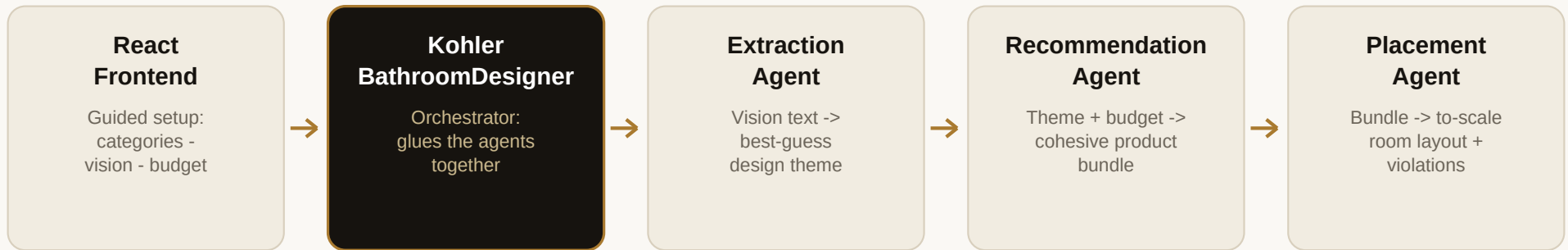
KOHLER design themes matched

**1,625**

scenarios swept with zero overlaps

## SYSTEM ARCHITECTURE

# One request, four cooperating agents



Only 5 categories (vanities, basins, toilets, showers, bathtubs) compete for floor space as spatial “anchors”; everything else (faucets, mirrors, lighting, bidet seats, cabinets, accessories) is attached by coordinate math to its owning fixture -- keeping the placement search small and fast while every product still ends up positioned correctly.

## KEY SPATIAL RULE

The fixture assembly runs along the room's actual LONGER wall -- not always the back wall -- so a 60"×96" room lays vanity - toilet - shower along its 96" run instead of being crammed into 60".

## TECH STACK

# Small, explainable, and fast to run

### Backend

#### Python + Flask + Flask-CORS

A lean REST surface for 5 focused endpoints -- no framework overhead.

### Frontend

#### React 18 + Vite

Fast HMR dev loop for a single guided conversational flow.

### “AI” layer

#### Deterministic weighted keyword / pattern matching

Reproducible and auditable; zero external LLM latency or cost.

### Data

#### Real KOHLER catalog (JSON, ~1,500 SKUs, scraped)

Real prices and product pages, not placeholder data.

### Visualization

#### Hand-rolled isometric + plan SVG

Renders straight from real placement coordinates -- never a staged mockup.

### Testing

#### Custom Python harnesses + a 1,625-scenario sweep

Placement correctness is safety-critical to the UX, so it's the most tested layer.

# Design tools usually show you a pretty render. This one tells you the truth about your room.

- **Honesty over illusion.**  
Most bathroom visualizers render a staged image; this one computes a real, to-scale placement and says so explicitly when a room genuinely can't fit the requested bundle -- no silent overlaps, no hidden compromises.
- **Spatial reasoning, not a template.**  
The placement engine reasons about which wall is actually longer, where the door swings, and what must physically touch what (faucet on the vanity, bidet on the toilet) -- the same rules a real installer would follow.
- **An agent that commits to an answer.**  
The style-matching step never dead-ends into "couldn't match, you pick" -- it always gives its best read on your taste, in plain language, while still leaving the final call to you.
- **Grounded in a real catalog.**  
Every recommendation is a real, purchasable KOHLER product with a real price and a real product page -- the bundle you see is the bundle you could actually buy.

[github.com/NikeBhide/NikhilBhideKohler](https://github.com/NikeBhide/NikhilBhideKohler)